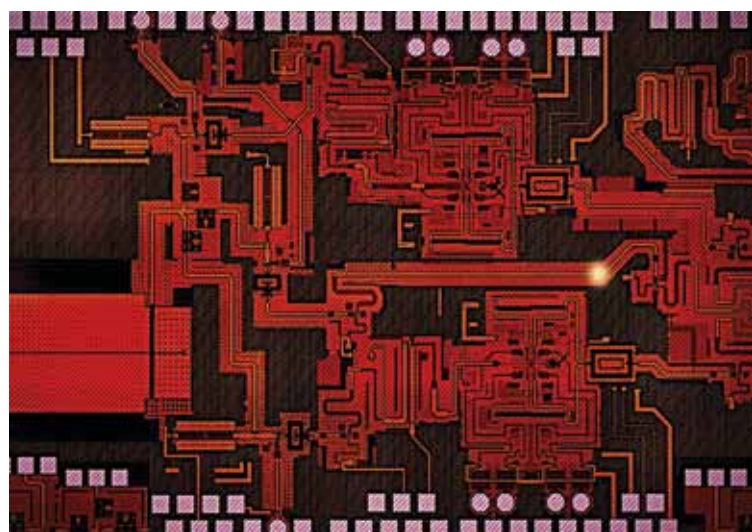


Autonomous robots for work in hazardous places

A University of Texas at Dallas research group has developed an autonomous robotic team of devices that can be used at hazardous or difficult-to-reach sites to make surveys and collect data, providing more and faster insights than human beings can deliver. The autonomous robotic team can rapidly learn the characteristics of environments it has never seen before. This allows rapid acquisition of holistic data by coordinated robotic teams to make data-driven decisions. The approach allows for more real-time data to be gathered rapidly and for streamlined software updates for the machines. The multi-robot, multi-sensor team includes a robot boat that carries sensors to measure water composition, as well as sonar to track objects below the surface and to provide aquatic remote sensing. At the same time, an overflying aerial robot collects hyperspectral images, providing an entire spectrum for every pixel in the image.



Aerial robot that can carry several cameras, an array of onboard sensors and a downwelling irradiance spectrometer—all used to gather data for mapping and to learn the characteristics of environments (Credit: <https://news.utdallas.edu>)



A data transfer system that pairs high-frequency silicon chips with a polymer cable as thin as a strand of hair (Credit: <https://news.mit.edu/2021>)

Silicon chips for rapid data transfer using hair-width cables

Researchers at the Massachusetts Institute of Technology (MIT) have developed a data transfer system that can transmit information ten times faster than a USB. The new link pairs high-frequency silicon chips with a polymer cable as thin as a hair strand. The system may one day boost energy efficiency in data centres and lighten the loads of electronics-rich spacecraft. The new link draws on benefits of both copper and fibre-optic conduits made of plastic polymer, so it is lighter and potentially cheaper to manufacture than traditional copper cables. But when the polymer link is operated with sub-terahertz electromagnetic signals, it is far more energy-efficient than copper in transmitting a high data load.

Automated sensor network for greenhouse gas emissions

Claimed to be the world's first, a fully automated sensor network for measuring urban greenhouse gas emissions using ground-based remote sensing of the atmosphere has been developed by scientists at the TUM Department of Electrical and Computer Engineering of the Technical University of

Munich (TUM). The sensor network called Munich Urban Carbon Column network (MUCCnet) consists of five high-precision optical instruments that analyse the sun's light spectra. They measure the concentration of carbon dioxide (CO₂), methane (CH₄), and carbon monoxide (CO). The sen-